EPFL

NII Inter-University Research Institute Corporation Research Organization of Information and Systems National Institute of Informatics

THESIS DISCUSSION · PROF. PRENDINGER

What should M2 learn to call a good trade?

Triple-barrier labels across 20 assets:
volatility, direction, and a controlled target experiment.

SCOPE	SecondaryModel's own 20 assets · daily candles · 7-day maximum holding period
EVIDENCE	Training diagnostics · verified M1 baseline · Pablo's saved M2 reference
PREPARED	22 October 2026 · short discussion for next week
UPDATE	28 October 2026 · UP M1 Sharpe −2.5744 · new M2 comparison pending

Which dates belong to which results?

Configured cutoffs · daily horizon = 7 bars · all dates below use YYYY-MM-DD.

Configuration	Data start	Train ends	Validation ends	Data ends
Earlier trial · separate PDF	2024-07-01	2025-05-30	2025-10-01	2026-01-25
Pablo GPU8 · now copied	2024-07-01	2025-08-31	2025-12-31	2026-08-01

Actual sample dates · after context and split rules

New daily inputs · Pablo's configuration
Train: 2024-06-30 – 2025-08-31
Val-Cal: 2025-09-08 – 2025-10-18
Val-Opt: 2025-10-26 – 2025-12-31
Test: 2026-01-08 – 2026-07-18

Other evidence and historical context
Label diagnostics: 2024-07-01 – 2025-08-24
Historical AG / Kumo Test · separate PDF:
2025-10-09 – 2026-01-18
Separate LightGBM Test:
2026-01-01 – 2026-09-23

The original loader creates a first training window on **2024-06-30**, one day before the configured start. Its context and embargo conventions are preserved. The realised Test dates match Pablo's saved reports.

GPU8 config observed 2026-10-27; daily input checks 2026-10-28; saved earlier trial, diagnostics and benchmark dates. Full LightGBM train/validation ranges accompany its result slides.

VERIFIED ON 28 OCTOBER 2026 · M1 BASELINE

The reproduced UP baseline has Sharpe −2.5744

Test: 2026-01-08 – 2026-07-18 · spread equity · 7-day horizon · fee 0.2%.

M1 side	Return	Sharpe	Max drawdown	Trades
UP	−54.25%	**−2.5744**	−57.99%	1045
DOWN	30.44%	1.7423	−17.04%	2580

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Target / direction baseline checks passed

27

Weekly return observations from 192 daily bars

−39.86%

Buy-and-hold return on the same Test period

Buy-and-hold also lost money in this period. All six targets share M1's Test trades. **New AutoGluon / Kumo comparison: pending**, 0 / 24 fits completed.

Independent daily data preparation reproduces saved GPU8 M1 metrics within 0.0001; existing UP multigranularity cache also checked. Spread equity smooths trade returns; Sharpe uses ×√(365.25/7). The inspected Test is exploratory.

Return and drawdown explain the “−45%” figure

Same 1045 UP trades · 2026-01-08 – 2026-07-18 · different equity calculations.

Equity calculation	Total return	Max drawdown	Sharpe
Spread · original baseline	−54.25%	−57.99%	−2.57
Cash · open positions at cost	−43.60%	−45.35%	−2.79
Cash · current market prices	−43.66%	−48.75%	−2.14

Saved cash report

−43.60% is the endpoint loss.
−45.35% is the worst fall from a prior peak.
Open positions remain at entry value.

Current cash code

Open positions follow current closing prices.
Interim losses become visible: **−48.75% drawdown**.
Use the same valuation rule for comparisons.

Replay of Pablo’s saved UP trade CSV on 2026-10-28; cash exposure 0.10 in both rows. Saved cash metrics match positions held at cost. Final-day entries can remain open: cash figures are endpoint equity, including the current marking fee.

Pablo’s saved UP M2 Sharpes are also negative

Test entries: 2026-01-08 – 2026-07-18 · 1d · TP labels · spread-equity engine.

Strategy	UP return	UP Sharpe	DOWN return	DOWN Sharpe
M1 · all calls	−54.25%	**−2.57**	30.44%	1.74
AutoGluon	−54.22%	**−2.36**	22.36%	1.58
CTTS	−58.65%	**−2.23**	23.36%	1.45
TabPFN	−21.50%	**−1.07**	20.70%	1.70
TabICL	−57.91%	**−2.43**	35.95%	1.91

> These are **Pablo’s saved TP results**. The new AutoGluon / Kumo target matrix uses the same dates and a verified M1 baseline; its M2 results are **pending**.

Saved GPU8 outputs fetched 2026-10-27. Replayed 8 runs (M1 and M2): 48 metrics agree within 0.0001; CSV returns have 6 decimals. 27 weekly horizon returns. Classifiers have not been refit to reproduce these outputs.

Inspecting the labels does not require retraining M2

Past candles
Market features + frozen M1
M1 proposes UP or DOWN.

→

Observed future candles
Apply the chosen exit rule.
Compute the net outcome.

→

Meta-label z
1 if the proposal paid;
0 otherwise.

WHAT WE CAN SHOW NOW

No retraining

Compare volatility, barrier touches and label changes using the same historical M1 calls.

WHAT A PERFORMANCE CLAIM NEEDS

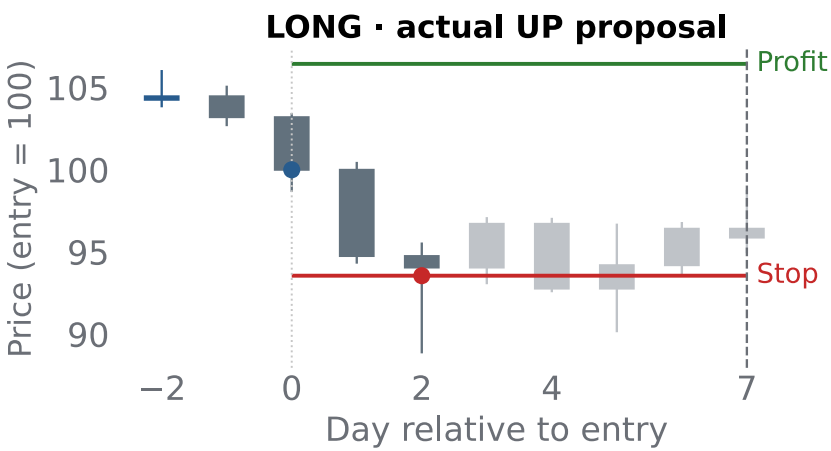
Retrain the secondary model

To test whether M2 learns the new target better, fit it on the new labels. Reuse M1's forecasts.

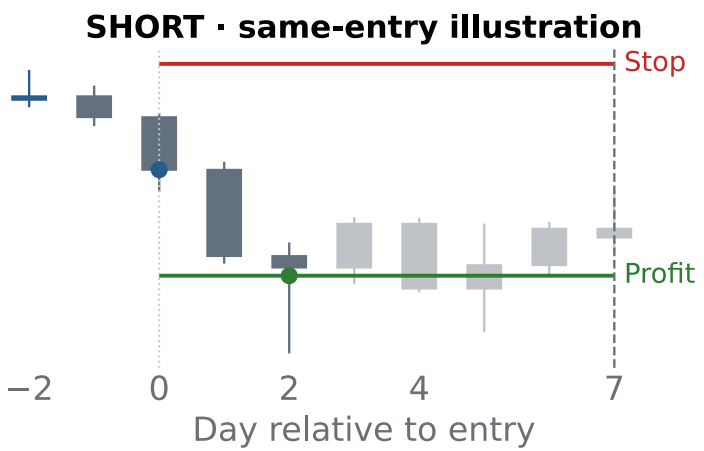
AFML §§3.4–3.7. Entry dates: 2024-07-01 – 2025-08-24. 7,593 eligible UP/DOWN proposals from 19 of SecondaryModel's 20 assets; TON unavailable in this parquet.

The same price move can stop a long and profit a short

BTC · 2024-07-03 · entry at day-0 close · symmetric k = 1 barriers.



Exit day 2 · net -6.6% · meta-label 0



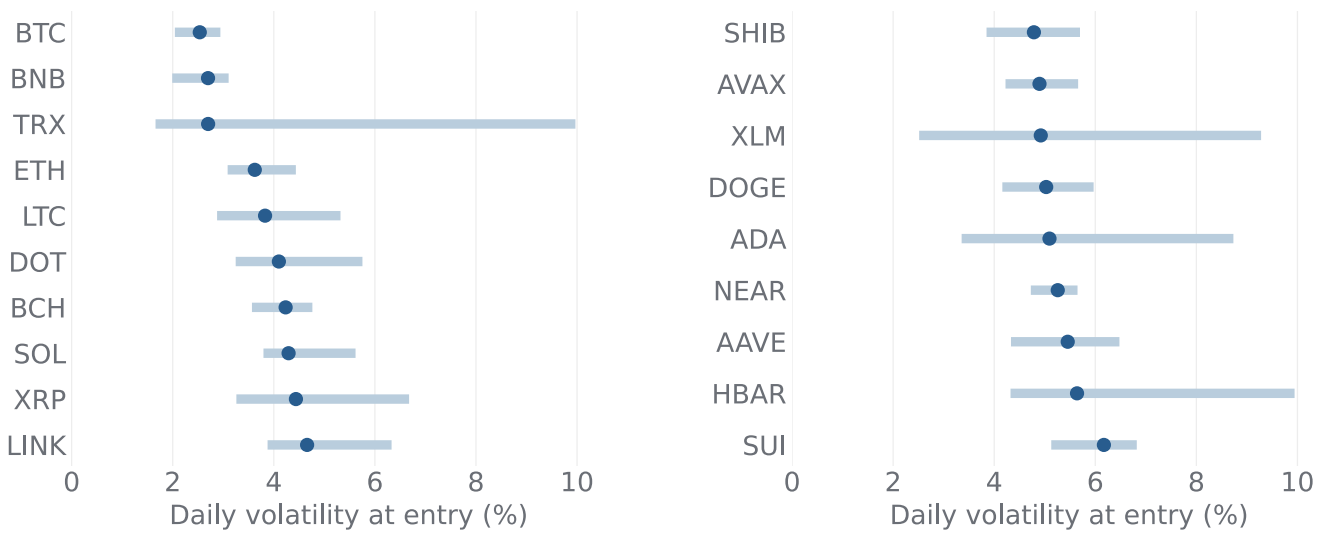
Exit day 2 · net +6.2% · meta-label 1

UP / long: profit above entry, stop below. DOWN / short: profit below, stop above.
The same entry-time volatility is used for both directions.

Actual training-period candles; explanatory episode selected for a stop. SHORT is a counterfactual, not a second Kronos call. Cost 0.2%; assumed barrier fills.

One common k still gives a different width at each entry

$$w_{a,t} = k\sigma_{a,t}\sqrt{7}$$

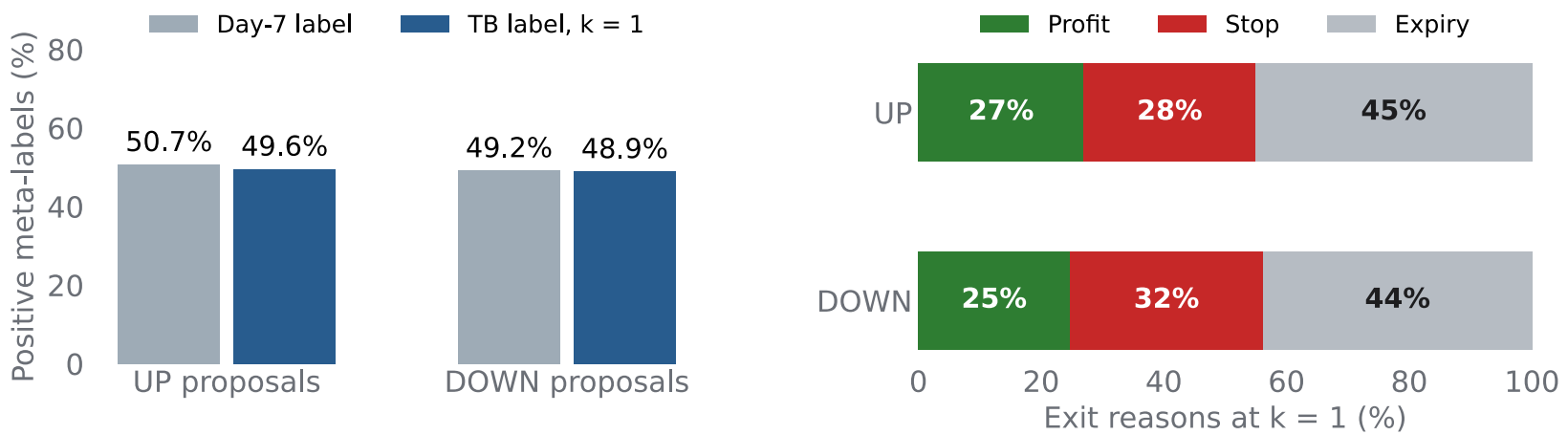


At k = 1, median widths are **6.7% for BTC**, **9.6% for ETH**, and **16.3% for SUI**.
σ changes with the asset and date; the chosen barrier is then fixed for that event.

Entry dates: 2024-07-01 – 2025-08-24. EWMA(100), known at entry; dots = median, lines = entry-date 10th–90th percentiles; √7 scaling.

Apply TBM to both sides; report the outcomes separately

1,921 UP proposals · 5,672 DOWN proposals · same k = 1 and same volatility estimator.

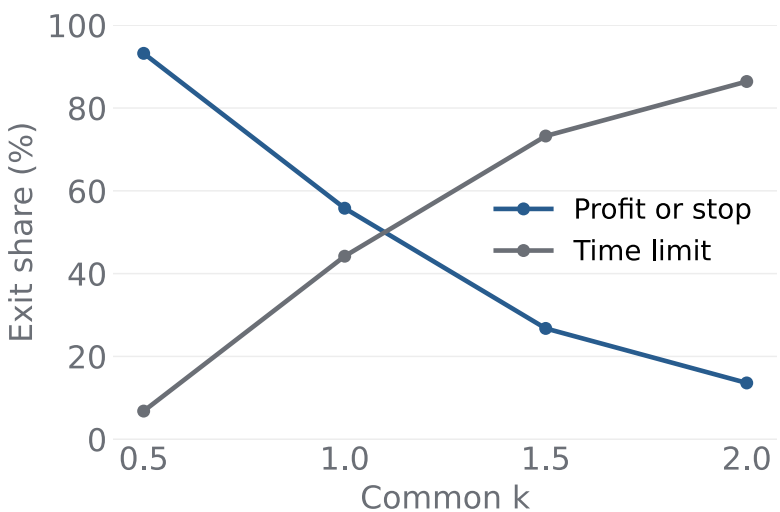


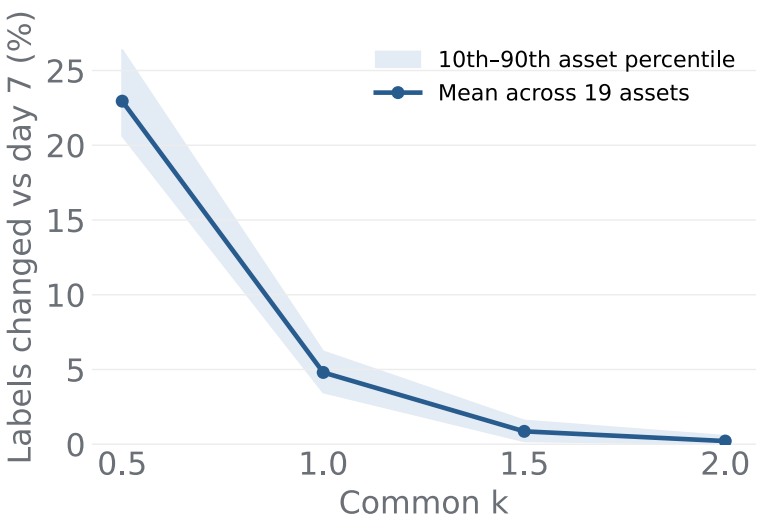
Meta-labels change for **5.9% of UP calls** and **4.4% of DOWN calls**.
These groups contain different dates and market conditions; this is a descriptive comparison.

Entry dates: 2024-07-01 – 2025-08-24. 19 of 20 assets; high/low touches, same-bar double touch = stop; cost 0.2%. Positive-label share is not M2's predictive accuracy.

A similar binary label can hide a different trade

Small k: closer exits. Large k: more trades remain open until day 7.





At k = 1: **55.8% horizontal exits**, but only **4.8% changed labels**, averaged across assets. The exit, return and holding time can change while “profitable / unprofitable” stays the same.

Entry dates: 2024-07-01 – 2025-08-24. Equal asset weights. Shading: 10th–90th percentiles across assets. Training diagnostics; these charts do not select an optimal k.

Start with a global k; select it for the intended trade

Start with one shared parameter

- Same k for all 20 assets and both sides.
- Volatility already adapts the width.
- Small grid: 0.5, 1, 1.5, 2.
- Keep the horizon, features and cost model fixed.

Choose on validation data

- Specify the intended exit policy first.
- Rank candidates by a predefined net portfolio objective.
- Check drawdown, coverage and stability across assets and time.
- Confirm the choice on fresh data.

There is no universal “ideal k”. Add separate direction or asset parameters only if they bring a stable validation benefit. A 50/50 label balance is not an optimization objective.

Proposed research protocol. AFML ch. 7 and ch. 11; López de Prado’s backtest-overfitting work. Existing 2026 test results have already been inspected.

CURRENT EXPERIMENT · PABLO'S DATES

The new M2 target comparison is still pending

Test: 2026-01-08 – 2026-07-18 · AutoGluon / Kumo · fixed day-7 execution.

UP M1 Sharpe: −2.5744
Verified baseline · return −54.25%

DOWN M1 Sharpe: 1.7423
Verified baseline · return 30.44%

M2 fit	TP	TB ref	k = 0.5	k = 1	k = 1.5	k = 2
AutoGluon · UP	Pending	Pending	Pending	Pending	Pending	Pending
AutoGluon · DOWN	Pending	Pending	Pending	Pending	Pending	Pending
Kumo · UP	Pending	Pending	Pending	Pending	Pending	Pending
Kumo · DOWN	Pending	Pending	Pending	Pending	Pending	Pending

0 / 24 new fits completed at this update. The earlier positive-UP target matrix is preserved in the separate historical PDF, using Test dates 2025-10-09–2026-01-18.

Status checked 2026-10-28. First submission stopped during preparation; the checker is fixed and the rerun is queued. Matched TB ref isolates label changes; all six targets share the verified M1 Test trade set.